

Neural Network Approach for Predicting Solar-Induced Fluorescence Using Synthetic Dataset

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Abstract

Solar-Induced Fluorescence (SIF) is a crucial indicator of plant photosynthetic activity and overall ecosystem health. Predicting SIF values enables better monitoring of vegetation productivity, stress, and carbon exchange, offering valuable insights for climate science, agriculture, and environmental management. With the upcoming launch of the ESA’s Fluorescence Explorer (FLEX) mission, the ability to predict and interpret SIF data is more important than ever. FLEX will provide unprecedented global SIF measurements, enhancing our understanding of plant physiology at multiple scales. The goal of this report is show the methodology employed for estimating SIF using a self-supervised approach and a dataset generated by combining the simulation by two radiative transfer models such as SCOPE and MODTRAN.

1 Introduction

Solar-induced fluorescence (SIF) has emerged as a vital biophysical indicator for assessing photosynthetic activity and vegetation health over diverse spatial scales. With its strong link to the internal dynamics of plant photosystems, SIF provides critical insights into ecosystem functioning, carbon cycling, and climate variability. Traditional retrieval methods have leveraged hyperspectral data and physics-based radiative transfer models (RTMs) to extract SIF signals, yet the inversion of the at-sensor radiance into meaningful biophysical parameters remains challenging. Recent advances have demonstrated the potential of neural network approaches to improve SIF estimation by integrating physical modeling principles with self-supervised learning strategies. In particular, methods such as those presented by Buffat et al. [1] have successfully combined spectral fitting techniques with deep learning frameworks to reconstruct SIF without relying on extensive ground truth measurements. These approaches exploit self-supervised loss functions that compare a forward-simulated at-sensor radiance, derived from a physical RTM, with the observed signal, thereby optimizing model parameters in a data-driven yet physically consistent manner. In contrast to previous studies that typically rely on a single RTM to simulate the at-sensor radiance, our work adopts a novel strategy by incorporating data generated from two distinct RTM models, SCOPE

and MODTRAN. This dual-model approach aims to capture a broader range of atmospheric and surface variability, ultimately enhancing the robustness and accuracy of the SIF predictions. By leveraging the complementary strengths of SCOPE’s detailed representation of plant-canopy processes and MODTRAN’s comprehensive treatment of atmospheric radiative transfer, our neural network is trained in a self-supervised manner to disentangle the complex interplay of reflectance, fluorescence, and atmospheric contributions in the observed signal. This paper details our methodology, discusses the integration of dual RTM-generated datasets into the training regime, and evaluates the performance of our neural network-based SIF estimation framework.

2 Dataset

Unlike Buffat et al., who relied on the HyPlant dataset for their SIF retrieval study, we did not have access to those in-situ measurements. Instead, we opted to generate a comprehensive synthetic dataset. In our approach, we used the SCOPE radiative transfer model to simulate reflectance and fluorescence signals by varying key parameters across realistic ranges. This allowed us to capture a wide spectrum of vegetation conditions and canopy processes. Next, we computed the corresponding at-sensor radiance by integrating these signals with an atmospheric

component generated using MODTRAN. This dual-model strategy not only replicates the full physical process—from plant reflectance and fluorescence to atmospheric modulation—but also provides a controlled environment to systematically assess and refine our neural network’s performance. Both MODTRAN and SCOPE have their input parameters, which we carefully changed to increase the variability of the data. Even though they are different models used for different kind of application, some parameters are present in both models. For this reason, when a simulation was performed in SCOPE, using a particular parameter shared between the two models, the same parameter was set equal also in MODTRAN, in order to avoid inconsistent data when the data of the two simulation was used to derive the radiance using equation 1.

$$LTOA \approx t_1 \cdot t_2 + \frac{t_1 \cdot (t_8 \cdot R + t_9 \cdot R + t_{10} \cdot R + t_{11})}{1 - t_3 \cdot R} + \frac{R + t_6 \cdot F + t_7 \cdot F}{1 - t_3 \cdot R} \quad (1)$$

The synthetic dataset utilized in the development of this project comprised 28.800 simulations, derived from 1.440 SCOPE simulations, each of which was combined with 20 distinct ambient conditions simulated using MODTRAN. Each simulation was generated within a defined spectral range of [650, 850] nm, using a very high spectral resolution of 0.1 cm^{-1} for computing the MODTRAN simulations.

2.1 SCOPE

SCOPE (Soil Canopy Observation of Photosynthesis and Energy fluxes) is a process-based radiative transfer model designed to simulate the interaction of light with vegetation canopies and underlying soil. It integrates key biophysical processes such as photosynthesis, energy balance, and fluorescence emission to provide detailed simulations of canopy reflectance and SIF signals. To generate variability in our synthetic dataset originating from the surface and vegetation, we systematically varied several key SCOPE input parameters using the ranges or specific values listed in Table 1. These chosen parameters and their ranges were partly inspired by the work of Scodellaro et al [2]. Other SCOPE parameters, such as carotenoid content (‘Cca’), dry matter (‘Cdm’), leaf structure (‘N’), and viewing geometry (‘tto’, ‘psi’), were kept constant at plausible values (e.g., ‘Cca=10.0’, ‘tto=0.0’, ‘psi=0.0’) throughout the simulations. The primary outputs generated

by these SCOPE runs were the spectral surface reflectance (the R term in Equation 1) and the top-of-canopy fluorescence emission spectrum (the F term in Equation 1).

Parameter	Values Simulated
Cab ($\mu\text{g}/\text{cm}^2$)	20.0, 25.0, 30.0, 40.0, 50.0
LAI (-)	0.0, 2.0, 3.0, 5.0, 8.0
Ta ($^{\circ}\text{C}$)	20.0, 30.0
fqe (-)	0.01, 0.03, 0.06, 0.1, 0.2, 0.5, 1.0, 2.0
tts (SZA) ($^{\circ}$)	0.0, 35.0, 60.0

Table 1: SCOPE parameters varied during synthetic data generation.

2.2 MODTRAN

MODTRAN (MODerate resolution atmospheric TRANsmission) is a comprehensive atmospheric radiative transfer model used to simulate the propagation of electromagnetic radiation through the Earth’s atmosphere. It computes the effects of absorption, scattering, and emission by atmospheric constituents, providing the necessary atmospheric transfer functions (the t_x terms in Equation 1) to convert surface-leaving radiance into at-sensor radiance (LTOA).

In our approach, each SCOPE simulation (yielding a specific R and F) was paired with multiple MODTRAN simulations representing different atmospheric states. To ensure consistency, key parameters shared between the models were synchronized: specifically, the Solar Zenith Angle (‘SZA’, corresponding to ‘tts’ in SCOPE) and the near-surface air temperature (‘TPTEMP’, corresponding to ‘Ta’ in SCOPE) used in a given MODTRAN run were inherited from the paired SCOPE simulation.

The primary source of variation introduced specifically during the MODTRAN stage came from simulating 20 distinct atmospheric scenarios, defined in the ‘ambient_conditions’ list within our processing script. These scenarios systematically altered parameters such as:

- Aerosol properties: Aerosol Optical Thickness (‘AOT’, e.g., ranging from 0.08 to 0.2) and implicitly Single Scattering Albedo (‘SSA’) and Asymmetry Factor (‘G’) via defaults.
- Trace gases: Ozone (‘O3’), Nitrogen Dioxide (‘NO2’), Carbon Monoxide (‘CO’) concentrations.

- Water vapor column (‘H2OSTR’, e.g., ranging from 1.5 to 2.5 cm).
- Cloud properties: Cloud fraction and cloud optical depth for several scenarios representing clear sky to moderate cloud cover.

These variations allowed us to generate diverse sets of atmospheric transfer functions ($t_1...t_{11}$) corresponding to each surface condition simulated by SCOPE. Other MODTRAN parameters related to geometry were kept constant across these atmospheric scenarios; notably, the ground altitude (‘GNDALT’) was fixed at 1 km, and the sensor altitude (‘H1ALT’) was set to 500 m. The standard mid-latitude summer atmospheric model (‘MODEL=2’) and a default visibility (‘VIS=23’ km) were used unless implicitly modified by the aerosol settings. Table 2 summarizes the key atmospheric component variations.

Component	Variations Introduced
Aerosols (AOT)	Varied from low (0.08) to higher (0.2) levels.
Trace Gases	Varied concentrations of O3, NO2, CO representing different pollution levels.
Water Vapor (H ₂ O)	Column amount varied (e.g., 1.5 cm, 2.0 cm, 2.5 cm).
Clouds	Scenarios included clear sky, slight cloud cover, and moderate cloud cover (varying fraction and optical depth).
Geometry	SZA and Ta inherited from SCOPE; GNDALT (1 km) and Sensor Alt (500 m) fixed.

Table 2: Summary of atmospheric variations simulated using MODTRAN via ‘ambient_conditions’ scenarios.

3 Architecture

The architecture we designed for estimating SIF is built upon a neural network framework that combines physical modeling principles with a self-supervised learning approach. The motivation behind this architecture is to effectively decompose at-sensor radiance into its constituent components—surface re-

flectance, fluorescence emission, and atmospheric effects—without relying on direct ground-truth SIF measurements. By leveraging synthetic data generated using the SCOPE and MODTRAN radiative transfer models, the network learns to estimate SIF in a physically consistent manner, ensuring robustness across varying environmental and observational conditions.

3.1 Input

The input data is not processed at the simulation level but instead grouped into small spatial patches, where each simulation represents a pixel, before being fed into the neural network. This patch-based approach allows the model to take advantage of local spatial consistency while reducing noise and improving the stability of the retrieval process. Instead of treating each pixel in isolation, which can introduce artifacts due to random spectral variations, the network considers small clusters of pixels together, helping it better disentangle reflectance, fluorescence, and atmospheric contributions.

Each patch consists of a fixed number of pixels arranged in a square window, for example 17×17 pixels, covering a localized area of the simulated scene. This size is chosen as a balance between two competing factors: ensuring that the patch contains enough spatial information for meaningful estimation while keeping the computational complexity manageable. If the patches were too small, the network would struggle to capture spatial patterns in the atmospheric and reflectance properties, making the SIF retrieval more sensitive to pixel-level noise. On the other hand, patches that are too large would increase memory requirements and computational costs without providing significant additional benefits.

3.2 Encoder

At the core of this framework lies an encoder-decoder neural network. The encoder is responsible for extracting meaningful features from the hyperspectral input data, which consists of simulated radiance measurements spanning the O2-A absorption band. Since hyperspectral data is inherently high-dimensional, the encoder first applies a dimensionality reduction step to condense the spectral information into a lower-dimensional latent representation. This is achieved through fully connected layers with non-linear activation functions, ensuring that the most relevant spectral features are retained while filtering out noise and redundant information. The encoder also takes into account auxiliary variables

such as solar zenith angle, observation geometry, and sensor altitude, which provide additional context for estimating SIF under different conditions.

3.3 Simulation

The simulation step is where the network “recreates” the at-sensor radiance based on its predicted physical parameters. In this step, the network takes its estimated values for things like surface reflectance, top-of-canopy fluorescence, and the atmospheric transfer functions, and feeds them into a forward physical model. This model is based on a four-stream radiative transfer formulation (derived from the SCOPE model) that captures both direct and diffuse fluxes in the O2-A absorption band. Naturally, the model used for recreates the at-sensor radiance is the same as the one used for creating the synthetic dataset, which is reported in 1. This approach allows the model to learn the underlying physical processes in a self-supervised manner, without requiring explicit SIF ground truth for every sample.

3.4 Loss

The loss function is the key element that drives the training of the network. It quantifies the error between the simulated at-sensor radiance (produced from the network’s predicted physical parameters via the forward simulation model) and the actual measured radiance from the physical sensor. In other words, it provides a numerical score indicating how close the simulation is to the real observation.

4 Differences with the original Architecture

The entire project is based on the network architecture proposed by Buffat et al [1], but some differences were needed to adapt the architecture to our need. In other words, we took some implementation choices that led us to diverge from applying the same solution used in the original paper. In the following lines, we are gonna cover the main differences between our implementation and their’s, highlighting the reasons behind our technical choices.

4.1 Physical Model

The major difference in our implementation compared to the one in the original paper is the use of a different physical model to reconstruct the radiance at the sensor. In the original paper, the radiance was

computed using equation 2 which considered also considered the solar irradiance normalized for the cosine of the solar zenith angle as term in the expression. In our case, the radiance was computed using equation 1, which does not consider this value. For this reason, we didn’t have to provide the value of the solar irradiance as input to the network, since in the original paper this value was used in the simulation step to perform the reconstruction of the input signal.

$$L_{at-s} = \frac{E_s^0 \cos(\theta_s)}{\pi} \times \left(t_1 + t_7 R + \frac{t_8 R + \bar{R} R t_{12} + \bar{R}(t_9 + t_{10})}{1 - t_2 \bar{R}} \right) + \left(t_5 f_s + \frac{\bar{f}(t_6 + t_{11} R)}{1 - t_2 \bar{R}} \right) \quad (2)$$

4.2 Encoder

The original architecture described by Buffat et al. [1] utilized an input normalization layer, a dimensionality reduction step (via a linear layer and ReLU), followed by an encoder module implemented as a multi-layer perceptron (MLP) with residual connections.

In contrast, for our single-pixel input vectors (LTOA spectrum + auxiliary data), we implemented and evaluated two main encoder architectures:

- Encoder with Single Projection Layer:** This architecture (‘SFMNNEncoder’ in the codebase) first processes the input through a sequence of fully connected layers (Linear, BatchNorm1d, ReLU, Dropout). The final layer in the backbone then projects the extracted features directly into a very high-dimensional space corresponding to all output variables concatenated spectrally (dimension $num_variables \times out_dim = 11 \times 3620$). This large output vector is intended to be reshaped later to separate the individual physical variables. While inspired by dimensionality reduction concepts, this approach relies on a single large linear transformation to generate all outputs simultaneously from the final feature representation. See table 3
- Encoder with Multi-Head Output:** This architecture (‘SFMNNEncoderWithHeadsSingle’ in the codebase), which was used for generating the results presented in this report, employs a shared backbone similar to the first approach, consisting of multiple fully connected layers with BatchNorm1d, ReLU activation, and Dropout

for feature extraction and regularization. However, instead of a single large projection, the final shared feature vector (of dimension ‘encoded_dim’, e.g., 8192) is fed into multiple, parallel output heads. Each head is a separate linear layer (‘nn.Linear(encoded_dim, out_dim)’ responsible for predicting one specific physical variable (one of the 9 required t variables, R , or F) across all spectral channels (‘out_dim’, e.g., 3620). A ReLU activation is applied to each head’s output to ensure non-negativity. This multi-head design allows each output stream to potentially specialize in mapping the shared features to its specific target variable, while still benefiting from the common feature extraction performed by the shared backbone. The final output shape is ($BatchSize$, 11, out_dim). See table 4

Component	Layer/Module	Parameters
14*6 Layers	BatchNorm1d	num_features=3623, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True
	(0) Linear	in_features=3623, out_features=8192, bias=True
	(1) BatchNorm1d	num_features=8192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True
	(2) ReLU	-
	(3) Dropout	p=0.1, inplace=False
	(4) Linear	in_features=8192, out_features=8192, bias=True
	(5) BatchNorm1d	num_features=8192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True
	(6) ReLU	-
	(7) Dropout	p=0.1, inplace=False
	(8) Linear	in_features=8192, out_features=8192, bias=True
	(9) BatchNorm1d	num_features=8192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True
	(10) ReLU	-
	(11) Linear	in_features=8192, out_features=8192, bias=True
	(12) BatchNorm1d	num_features=8192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True
	(13) ReLU	-
latent_proj	Linear	in_features=8192, out_features=39820, bias=True

Table 3: Architecture of SFMNNEncoder

Component	Layer/Module	Parameters
12*shared Layers	BatchNorm1d	num_features=3623, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True
	(0) Linear	in_features=3623, out_features=8192, bias=True
	(1) BatchNorm1d	num_features=8192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True
	(2) ReLU	-
	(3) Dropout	p=0.1, inplace=False
	(4) Linear	in_features=8192, out_features=8192, bias=True
	(5) BatchNorm1d	num_features=8192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True
	(6) ReLU	-
	(7) Dropout	p=0.1, inplace=False
	(8) Linear	in_features=8192, out_features=8192, bias=True
	(9) BatchNorm1d	num_features=8192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True
	(10) ReLU	-
	(11) Dropout	p=0.1, inplace=False
	(12) Linear	in_features=8192, out_features=3620, bias=True
heads	ModuleList	11 × Linear(in_features=8192, out_features=3620, bias=True)

Table 4: Architecture of SFMNNEncoderWithHeadsSingle

4.3 Number and Type of Encoded Variables

A crucial adaptation compared to the original work by Buffat et al. stemmed from our use of a different physical model for LTOA reconstruction (Equation 1). Their model (Equation 2) required retrieving 6 atmospheric variables along with reflectance and fluorescence. Our simplified model necessitates the

retrieval of 9 specific atmospheric transfer functions ($t_1, t_2, t_3, t_6, t_7, t_8, t_9, t_{10}, t_{11}$), in addition to the surface reflectance (R) and the fluorescence spectrum (F).

Therefore, the encoder, specifically the multi-head architecture used for obtaining our results, was designed with exactly 11 output heads (‘num_variables=11’). Each head produces a vector corresponding to one of these physical quantities across the relevant spectral dimension (‘out_dim’). The order of these heads in our implementation corresponds to ($t_1, t_2, t_3, t_6, t_7, t_8, t_9, t_{10}, t_{11}, R, F$). This output is then used within the training loop’s simulation step to reconstruct the LTOA signal via Equation 1 for comparison with the target LTOA in the loss function.

4.4 Loss

In the original paper, the loss function is a self-supervised objective that combines a primary reconstruction error with several regularizers to ensure physical consistency in the SIF retrieval process. Specifically, the main term is a mean squared error calculated over the 750–770 nm spectral range that measures the difference between the observed at-sensor radiance and the reconstructed radiance produced by a forward simulation model based on radiative transfer principles. To further guide the network in disentangling fluorescence, reflectance, and atmospheric effects, the loss includes additional regularizers: a fluorescence regularizer (l_f) that weights the reconstruction error in spectral regions with high fluorescence signal-to-noise ratio using a sensor-specific weight function derived from a linearized inversion approach, a physiological regularizer (l_{NDVI}) that enforces consistency with vegetation indices, and an atmospheric regularizer (l_{atm}) that ensures physically plausible atmospheric parameter estimates.

Based on the fact that the physical model used for reconstructing the radiance at-sensor was different from the one used in the original paper, we had to adapt the loss function accordingly.

We experimented with progressively more complex loss formulations to guide the learning process:

- **SFMNNLoss (Conceptual Baseline):** Our initial conceptual approach aimed to mirror the original paper [1] by using primarily the Mean Squared Error (MSE) between the target LTOA and the reconstructed LTOA within a specific spectral window (e.g., 650-850 nm). This forces the network to learn the non-linear mapping defined by Equation 1. Early experiments indicated this alone was insufficient for robust SIF

retrieval due to its small contribution to the total LTOA signal. (Note: The class ‘SFMNNLoss’ in the provided code includes additional terms and seems designed for patch-based input, differing from this baseline concept used for single-pixel models).

- **SFMNNLossEnhanced** (‘SFMNNLossSingle’ class): To address the poor learning signal for fluorescence (F) observed with the baseline approach, we implemented the ‘SFMNNLossSingle’ class. This loss function retains the core LTOA reconstruction MSE (‘recon_loss’) but enhances it with several terms inspired by the original paper and adapted for our single-pixel setup:

- *SIF MSE*: A direct MSE term comparing the full predicted fluorescence spectrum against the target fluorescence obtained from our synthetic dataset. This provides a strong, direct gradient for the fluorescence output head.
- *NDVI Penalty*: A physiological regularizer that penalizes the predicted SIF in pixels where the NDVI, calculated using the network’s predicted reflectance, is below a certain threshold. This encourages the network to suppress fluorescence predictions in non-vegetated or stressed areas.
- *SIF Regularization*: An L1 norm applied to the full predicted SIF spectrum, promoting sparsity or penalizing large fluorescence values overall (though its weight was often set to 0 in practice, since we noticed we experimented more with the other SIF focused term).
- *Signal-Weighted Loss*: An attempt to dynamically weight the LTOA reconstruction MSE based on the predicted SIF magnitude (often disabled by setting its weight to 0 due to potential instability).

This ‘SFMNNLossSingle’ class served as the primary loss function for the training loop focused on LTOA and F reconstruction without direct loss on the atmospheric t parameters.

- **PhysicsRegularizedLoss**: This final loss formulation extended ‘SFMNNLossSingle’ by incorporating direct MSE terms for all predictable physical components. In addition to the terms present in ‘SFMNNLossSingle’ (though potentially re-weighted), it adds:

- *Reflectance MSE*: Compares the predicted reflectance spectrum (R) against the target reflectance from the dataset.
- *Atmospheric MSE*: Compares the 9 predicted atmospheric spectra ($t_1, t_2, t_3, t_6 \dots t_{11}$) against their corresponding targets selected from the dataset.

This provides the most comprehensive supervision, giving direct feedback on the accuracy of each individual output head of the network.

It is important to note that the ability to include direct MSE terms on F , R , and the t variables relies entirely on using a synthetic dataset where these ground truth values are known. In a real-world scenario, only the LTOA reconstruction term (and potentially physics-based regularizers not requiring ground truth F , R , or t) could be used for self-supervision. Our experiments using the enhanced losses serve to diagnose the network’s learning capabilities for each component and potentially pre-train a model that could be fine-tuned on real data using a simpler, purely self-supervised LTOA reconstruction loss.

5 Results

The training process, monitored via the loss curves shown in Figure 1, revealed several key challenges in retrieving physical parameters from LTOA radiance using the neural network approach with our synthetic dataset.

Initial experiments using the basic ‘SFMNNLoss’ showed that the network could learn to reduce the overall loss over epochs, but struggled significantly with accurately predicting the fluorescence component (F), isolating the individual F , R , and t terms remained difficult.

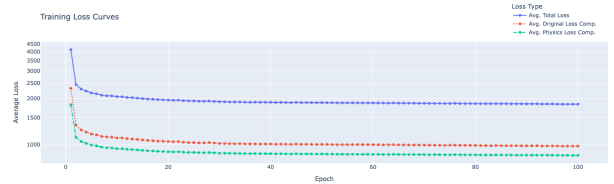


Figure 1: Example training loss curve for the multi-head encoder using the ‘PhysicsRegularizedLoss’. The loss decreases, indicating learning occurred.

Subsequent experiments incorporating direct MSE terms for F (‘SFMNNLossEnhanced’) and later t and R (‘PhysicsRegularizedLoss’) provided stronger gradients for these components, as illustrated by example

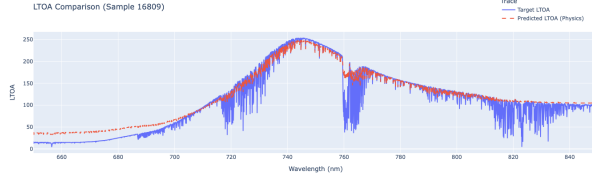


Figure 2: Example LTOA spectrum reconstruction. Solid line: Target LTOA. Dashed line: LTOA reconstructed by the network using its predicted components (trained with ‘PhysicsRegularizedLoss’).

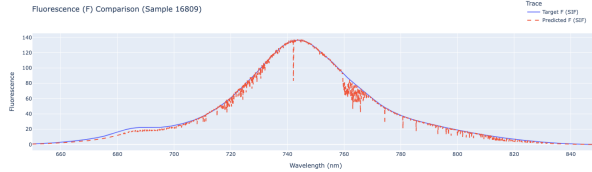


Figure 3: Example Fluorescence (F) prediction vs target. Solid line: Target F . Dashed line: Predicted F by the network (trained with ‘PhysicsRegularizedLoss’). Note the discrepancy, especially at lower target values.

reconstructions generated using enhanced loss functions (Figures 2, 3, 4, 5), even when LTOA reconstruction improved (Figure 2). This led to some improvements in reconstructing F , but primarily for training examples where the target fluorescence signal was already high (as qualitatively observed in plots like Figure 3). A persistent issue emerged: the network tended to overshoot the predicted F values, particularly for cases where the true fluorescence was low. This suggests that because the absolute contribution of F to the total LTOA signal is often small, the network prioritized matching high- F examples where the MSE on F was largest, leading to a systematic overestimation bias elsewhere.

Furthermore, these experiments reinforced the difficulty in forcing the network to significantly overfit the training data. This behavior – learning skewed towards high-signal examples, overshooting elsewhere, poor generalization, and resistance to overfitting – strongly suggests that the inverse problem itself, as formulated, is highly ill-posed. Theoretically, multiple distinct combinations of atmospheric parameters (t), reflectance (R), and fluorescence (F) might produce nearly identical LTOA spectra, given the limited input information. The network struggles to find a unique mapping and may learn an average or biased solution.

Compounding these issues was a limitation in the

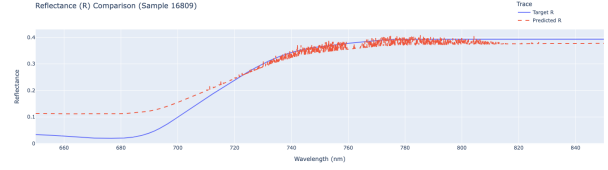


Figure 4: Example Reflectance (R) prediction vs target. Solid line: Target R . Dashed line: Predicted R by the network (trained with ‘PhysicsRegularizedLoss’).

synthetic dataset itself. The parameter ranges chosen for SCOPE and MODTRAN inadvertently included examples with potentially unrealistically high fluorescence values. As seen in Figure 3, the target fluorescence sometimes reaches high levels, which likely exacerbated the network’s tendency to focus on and overshoot the F component, as these outlier examples disproportionately influence the loss. Ensuring physically plausible target values across the entire dataset is critical.

Tuning the relative weights (λ) of the loss components proved sensitive and did not fully resolve these core issues. While potential minor issues in normalization or variable scaling might contribute, the dominant challenges appear to be the ill-posed nature of the retrieval and the distribution (including realism and outliers) of target values in the synthetic dataset.

6 Conclusion

This project aimed to adapt the self-supervised neural network architecture proposed by Buffat et al. [1] for SIF retrieval, utilizing a synthetic dataset generated by coupling the SCOPE and MODTRAN radiative transfer models and employing a simplified LTOA reconstruction formula (Equation 1). Key modifications included adapting the encoder outputs and experimenting with different network structures and loss function designs (‘SFMNNLoss’, ‘SFMNNLossEnhanced’, ‘PhysicsRegularizedLoss’).

Our results indicate that while the neural network framework shows potential for LTOA reconstruction (Figure 2), the accurate retrieval of underlying physical parameters, particularly SIF (F) (Figure 3), faces significant challenges. These stem from the likely ill-posed nature of the inverse problem with the current input data structure, leading to ambiguities where different parameter combinations yield similar LTOA signals. This manifested as difficulty in achieving good generalization and even in overfitting the training data (Figure 1).

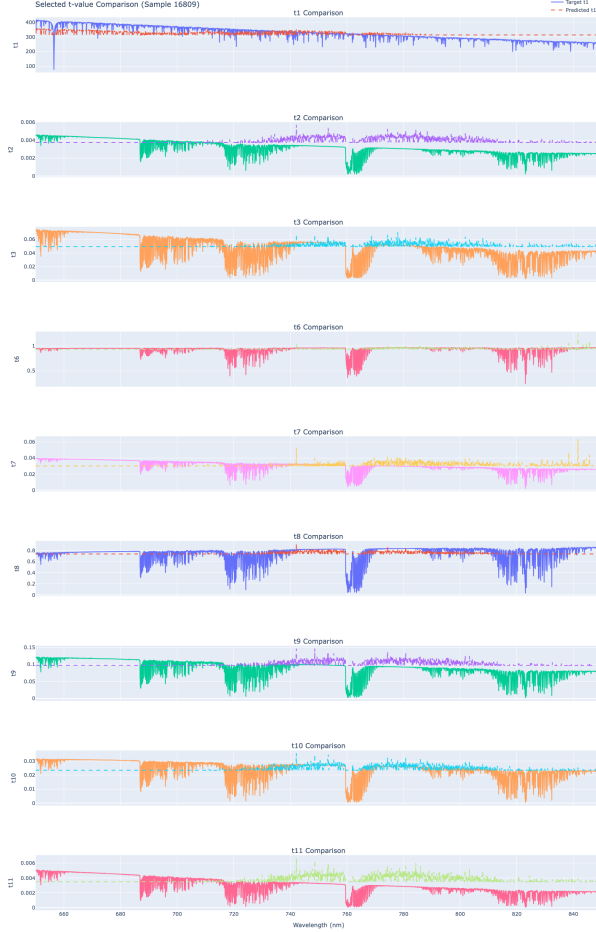


Figure 5: Example prediction vs target for selected atmospheric variables (e.g., t_1 , t_3 , t_8). Solid lines: Targets. Dashed lines: Predictions (trained with ‘PhysicsRegularizedLoss’).

Furthermore, the relatively small contribution of the fluorescence signal to the total LTOA radiance complicates loss optimization. Even with direct MSE loss terms on F , the network tended to perform well only on high-SIF examples while overshooting predictions for lower, more typical SIF values (observed in Figure 3). This bias was likely exacerbated by the presence of potentially unrealistically high fluorescence values in the generated synthetic dataset, highlighting a critical challenge in creating perfectly representative training data. The retrieval of reflectance (R) and atmospheric parameters (t) also showed discrepancies (Figures 4, 5).

The use of synthetic data, while providing ground truth, did not overcome these fundamental issues. Simply adding direct supervision via MSE terms was insufficient when the input features lack the neces-

sary constraints or when the target data distribution contains outliers or biases.

Future work should prioritize addressing these core challenges:

- **Dataset Realism and Balance:** Critically review and refine the SCOPE and MODTRAN parameter ranges (especially ‘fqc’) to ensure physically realistic target values. Investigate methods to balance the dataset or weight the loss to prevent outliers or low-signal regions from being ignored.
- **Enhancing Input Information:** Incorporating multi-angle observations, varied solar geometries, or auxiliary atmospheric data remains crucial for reducing retrieval ambiguity. Ideally, using a real informative dataset for validation or fine-tuning.
- **Advanced Loss Functions:** Explore loss functions beyond simple MSE, such as robust loss functions (e.g., Huber) or relative error metrics, less sensitive to outliers or signal magnitude. Investigate adaptive weighting schemes.
- **Refining Loss Function Weights:** Continue systematic hyperparameter optimization of the λ weights. This is a time-consuming process.
- **Architectural Exploration:** Consider architectures designed for ill-posed problems or incorporating stronger physics-based inductive biases.
- **Regularization Strategies:** Explore regularization targeting the physical plausibility of the outputs (t , R , F).

In conclusion, successfully applying self-supervised neural networks for SIF retrieval requires a careful balance between realistic data generation, sufficient input information to constrain the ill-posed problem, and robust loss function design. Our work underscores these complexities and provides direction for future improvements in synthetic data generation and model development for physically-based parameter retrieval.

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